

Project Description (Schema & Initial Data)

We have attached the relational schema and screenshots of the data tables of each relation after the initial SQL script is run.

RELATIONAL SCHEMA

MaintenanceRecord1(MaintenancePerformed: VARCHAR, NumberOfWorkers: INTEGER)

- Primary Key: MaintenancePerformed

MaintenanceRecord2(RideName: VARCHAR, RecordID: INTEGER, MaintenancePerformed: VARCHAR, MaintenanceDate: DATE)

- Primary Key: RideName, RecordID
- Foreign Key: RideName

**This colour & underline means partial key as google sheets does not allow dotted underlines.*

Ride(RideName: VARCHAR, Capacity: INTEGER, InstallDate: DATE)

- Primary Key: RideName

Ticket1(ValidFrom: DATE, ValidUntil: DATE, ValidHours: INTEGER)

- Primary Key: ValidFrom, ValidHours

Ticket2(TicketID: INTEGER, **BookingNumber**: INTEGER, **ValidFrom**: DATE, RemainingUses: INTEGER, **ValidHours**: INTEGER)

- Primary Key: TicketID
- Foreign Key: BookingNumber, ValidFrom, ValidHours

FastPassTicket(**TicketID**: INTEGER, NumberOfRides: INTEGER, ValidHours: INTEGER)

TicketID is foreign key

- Primary Key: TicketID
- Foreign Key: TicketID

Customer(CustomerID: INTEGER, CustomerName: VARCHAR, Sex: VARCHAR, DOB: DATE)

- Primary Key: CustomerID

Guest(**CustomerID**: INTEGER, DateOfVisit: DATE)

- Primary Key: CustomerID
- Foreign Key: CustomerID

LoyaltyMember(**CustomerID**: INTEGER, LoyaltyID: INTEGER, Points: INTEGER)

- Primary Key: CustomerID
- Candidate Key: LoyaltyID
- Foreign Key: CustomerID

Booking1(BookingDate: DATE, Cost: INTEGER)

- Primary Key: BookingDate

Booking2(BookingNumber: INTEGER, BookingDate: DATE, **CustomerID**: INTEGER)

- Primary Key: BookingNumber

- Foreign Key: CustomerID (NOT NULL)

HotelStay(HotelBookingID: INTEGER, NumGuests: INTEGER, CheckInDate: DATE, CheckOutDate: DATE, **HotelName**: VARCHAR, **RoomName**: VARCHAR, **BookingNumber**: INTEGER)

- Merged with BookedAt, AtRoomType, HasHotelStay
- Primary Key: HotelBookingID
- Foreign Key: HotelName (NOT NULL), RoomName (NOT NULL), BookingNumber (NOT NULL)

RoomType1(BaseRate: FLOAT, Category: VARCHAR)

- Primary Key: Category

RoomType2(RoomName: VARCHAR, MaxGuests: INTEGER, Category: VARCHAR)

- Primary Key: RoomName

Hotel1(City: VARCHAR, Province: VARCHAR, PostalCode: VARCHAR)

- Primary Key: PostalCode

Hotel2(HotelName: VARCHAR, StreetAddress: VARCHAR, PostalCode: VARCHAR)

- Primary Key: HotelName

RateModifier(RateCode: INTEGER, Modifier: FLOAT)

- Primary key: RateCode

SeasonPass1(SeasonPassLevel: VARCHAR, SeasonStart: DATE, SeasonEnd: DATE,)

- Primary Key: SeasonPassLevel, SeasonStart

SeasonPass2(UUID: INTEGER, SeasonPassLevel: VARCHAR, SeasonStart: DATE, **LoyaltyID**: INTEGER)

- Primary Key: UUID
- Foreign Key: LoyaltyID (NOT NULL, UNIQUE)

ForRide(**RideName**: VARCHAR,

TicketID: INTEGER)

- Primary Key: RideName, TicketID
- Foreign Key: RideName, TicketID

OffersRoomType(**HotelName**: VARCHAR,

RoomName: VARCHAR,

Quantity: INTEGER)

- Primary Key: HotelName, RoomName
- Foreign Key: HotelName, RoomName

BookedWithRateModifier(**RateCode**: INTEGER,

BookingNumber: INTEGER)

- Primary Key: RateCode, BookingNumber
- Foreign Key: RateCode, BookingNumber

SeasonPassSpecial(**UUID**: INTEGER,

RateCode: INTEGER)

- Primary Key: UUID, RateCode
- Foreign Key: UUID, RateCode

BOOKEDWITHRATEMODIFIER

RATECODE	BOOKINGNUMBER
0	1005
1	1001
2	1002
3	1003
4	1004

BOOKING1

BOOKINGDATE	COST
2025-07-10 00:00:00	120
2025-07-11 00:00:00	95
2025-07-12 00:00:00	90
2025-07-14 00:00:00	150
2025-07-16 00:00:00	110
2025-07-18 00:00:00	200
2025-08-01 00:00:00	130
2025-09-01 00:00:00	135
2025-09-14 00:00:00	140
2025-09-20 00:00:00	160

BOOKING2

BOOKINGNUMBER	BOOKINGDATE	CUSTOMERID
1001	2025-07-10 00:00:00	1
1002	2025-08-01 00:00:00	2
1003	2025-09-01 00:00:00	3
1004	2025-07-16 00:00:00	4
1005	2025-07-18 00:00:00	5
1111	2025-07-10 00:00:00	1
1112	2025-07-10 00:00:00	1
1113	2025-07-11 00:00:00	1
1114	2025-07-12 00:00:00	1
1115	2025-07-12 00:00:00	1
2223	2025-09-14 00:00:00	3
2224	2025-09-14 00:00:00	3
2225	2025-09-14 00:00:00	3
2226	2025-09-14 00:00:00	3
2227	2025-09-20 00:00:00	3
2323	2025-07-18 00:00:00	5
2424	2025-07-18 00:00:00	5
2525	2025-07-18 00:00:00	5
2626	2025-07-18 00:00:00	5
2727	2025-07-18 00:00:00	5
7433	2025-09-01 00:00:00	8
7434	2025-09-01 00:00:00	8
7435	2025-09-01 00:00:00	8
7436	2025-09-01 00:00:00	9
7437	2025-09-01 00:00:00	9
7438	2025-09-01 00:00:00	10

CUSTOMER

CUSTOMERID	CUSTOMERNAME	SEX	DOB
1	Alice Johnson	F	1990-05-14 00:00:00
2	Bob Smith	M	1988-08-09 00:00:00
3	Carol White	F	1995-11-02 00:00:00
4	David Brown	M	2000-01-25 00:00:00
5	Eve Davis	F	1993-03-07 00:00:00
6	John Guest	M	1930-06-14 00:00:00
7	John Guest II	M	1947-03-09 00:00:00
8	John Guest III	M	1965-10-02 00:00:00
9	John Guest IV	M	1995-04-25 00:00:00
10	Terry Margaret	F	1967-09-07 00:00:00
11	Sally Smith	F	1993-05-08 00:00:00
12	Patty White	F	1993-06-29 00:00:00
13	Aaron Davis	Unspecified	1985-08-07 00:00:00
14	Jessie Brown	Unspecified	1985-05-21 00:00:00
15	Quinn Jude	Unspecified	1985-04-17 00:00:00
16	River Quinn	Unspecified	1985-11-11 00:00:00
25	James Guest	M	1930-12-01 00:00:00
26	James Guest Jr.	M	1930-09-09 00:00:00
27	James Guest III	M	1930-10-22 00:00:00
28	James Guest IV	M	1995-04-24 00:00:00
29	James Margaret	M	1995-09-02 00:00:00
30	Jeff Guest	M	1995-06-11 00:00:00
31	Jeff Guest II	M	1949-06-17 00:00:00
32	Jeff Guest III	M	1949-10-16 00:00:00
33	Jeff Guest IV	M	1995-04-30 00:00:00
34	Jeff Margaret	M	1949-09-30 00:00:00
35	Tom Margaret	M	1949-09-30 00:00:00
100	James Vick	Non-binary	2000-03-27 00:00:00
101	Travis Albanza	Non-binary	2000-01-31 00:00:00
102	Ian Alexander	Non-binary	1994-02-14 00:00:00
103	Oilly Alexander	Non-binary	1973-12-12 00:00:00
226	Michael Todd	M	1930-06-14 00:00:00
227	Janice Mann	F	1947-03-09 00:00:00
228	Vanessa Jest	F	1965-10-02 00:00:00
229	Johnny Peterson Jr	M	1995-04-25 00:00:00
330	Mary Margaret	F	1967-09-07 00:00:00
336	Richard Guest	M	1930-06-14 00:00:00
337	Ricky Guest II	M	1947-03-09 00:00:00
338	Bobby Smith	M	1965-10-02 00:00:00
339	Bobbert Smith	M	1995-04-25 00:00:00
340	Tammy Lammy	F	1967-09-07 00:00:00
446	Michella Todd	F	1930-06-14 00:00:00
447	Jasmine Robertson	F	1947-03-09 00:00:00
448	Andy Jest	M	1965-10-02 00:00:00
449	Jonny Peters Jr	M	1995-04-25 00:00:00
556	Richardo Ramirez	M	1930-06-14 00:00:00

557	Richy Guest Sr	M	1947-03-09 00:00:00
558	Waldo Smith	M	1965-10-02 00:00:00
559	Danny Smiths	M	1995-04-25 00:00:00
666	Richardson Guest	M	1930-06-14 00:00:00
667	Ramos Guest II	M	1947-03-09 00:00:00
668	Bravado Smith	M	1965-10-02 00:00:00
669	Benjamin Smith	M	1995-04-25 00:00:00
776	Pamela Todd	F	1930-06-14 00:00:00
777	Jasmine Johnson	F	1947-03-09 00:00:00
778	Andrew Jest	M	1965-10-02 00:00:00
779	Jackson Peters Jr	M	1995-04-25 00:00:00
886	Richardoz Ramirez	M	1930-06-14 00:00:00
887	Raoni Guest Sr	M	1947-03-09 00:00:00
888	Woodrow Smith	M	1965-10-02 00:00:00
889	Wilson Smiths	M	1995-04-25 00:00:00
4440	Marlena Margaretta	F	1967-09-07 00:00:00
5550	Samson Ma	M	1967-09-07 00:00:00
6660	Rita Lammy	F	1967-09-07 00:00:00
7770	Arlana Margarette	F	1967-09-07 00:00:00
8880	Samsung Manson	M	1967-09-07 00:00:00

FASTPASSTICKET

TICKETID	NUMBEROFRIDES	VALIDHOURS
5001	10	8
5002	8	10
5003	5	6
5004	7	12
5005	3	4

FORRIDE

RIDENAME	TICKETID
BumperCars	5005
BumperCars	5115
BumperCars	5116
BumperCars	6670
BumperCars	8890
BumperCars	63207
FerrisWheel	5002
FerrisWheel	5112
FerrisWheel	6667
FerrisWheel	8887
FerrisWheel	63204
HauntedHouse	5003
HauntedHouse	5113
HauntedHouse	6668
HauntedHouse	8888
HauntedHouse	63205
LogFlume	3333
LogFlume	5004
LogFlume	5114
LogFlume	6669
LogFlume	8889
LogFlume	63206
LogFlume	63208
RollerCoaster	3334
RollerCoaster	5001
RollerCoaster	6666
RollerCoaster	8886
RollerCoaster	63203

GUEST

CUSTOMERID	DATEOFVISIT
6	2025-05-10 00:00:00
7	2025-06-11 00:00:00
8	2025-07-12 00:00:00
9	2023-07-09 00:00:00
10	2023-07-18 00:00:00
226	2024-08-14 00:00:00
227	2024-08-09 00:00:00
228	2024-08-02 00:00:00
229	2024-08-22 00:00:00
330	2024-08-07 00:00:00
336	2024-08-07 00:00:00
337	2024-08-07 00:00:00
338	2024-09-02 00:00:00
339	2024-09-25 00:00:00
340	2024-10-07 00:00:00
446	2023-10-14 00:00:00
447	2023-10-09 00:00:00
448	2023-10-02 00:00:00
449	2023-10-25 00:00:00
556	2022-11-14 00:00:00
557	2022-11-09 00:00:00
558	2022-11-02 00:00:00
559	2023-11-25 00:00:00
666	2023-12-14 00:00:00
667	2023-12-09 00:00:00
668	2023-12-02 00:00:00
669	2023-12-25 00:00:00
776	2024-01-14 00:00:00
777	2024-01-09 00:00:00
778	2024-01-02 00:00:00
779	2024-01-25 00:00:00
886	2022-02-14 00:00:00
887	2022-02-09 00:00:00
888	2022-02-02 00:00:00
889	2023-03-25 00:00:00
4440	2023-10-07 00:00:00
5550	2024-11-07 00:00:00
6660	2023-01-07 00:00:00
7770	2024-01-07 00:00:00
8880	2019-04-07 00:00:00

HOTEL1

POSTALCODE	PROVINCE	CITY
A1B2C3	BC	Vancouver
B2C3D4	AB	Calgary
C3D4E5	ON	Toronto
D4E5F6	QC	Montreal
E5F6G7	NS	Halifax

HOTEL2

HOTELNAME	MAXGUESTS	POSTALCODE
OceanView	200	A1B2C3
MountainInn	150	B2C3D4
CityCenter	300	C3D4E5
LakesideLodge	100	D4E5F6
HarborHotel	180	E5F6G7

HOTELSTAY

HOTELBOOKINGID	NUMGUESTS	CHECKINDATE	CHECKOUTDATE
1	2	2025-07-10 00:00:00	2025-07-12 00:00:00
2	4	2025-07-12 00:00:00	2025-07-15 00:00:00
3	3	2025-07-14 00:00:00	2025-07-17 00:00:00
4	2	2023-07-16 00:00:00	2023-07-18 00:00:00
5	5	2023-07-18 00:00:00	2023-07-21 00:00:00

HOTELNAME	ROOMNAME	BOOKINGNUMBER
OceanView	Single	1001
MountainInn	Double	1002
CityCenter	Suite	1003
LakesideLodge	Cabin	1004
HarborHotel	Penthouse	1005

LOYALTYMEMBER

CUSTOMERID	LOYALTYID	POINTS	UUID
1	101	120	10001
2	102	80	10002
3	103	200	10003
4	104	0	10004
5	105	30	10005
11	106	40	10006
12	107	10	10007
13	108	64	10008
14	109	30	10009
15	110	17	10010
16	111	2201	10011
25	116	1	20001
26	117	3	20002
27	118	4	20003
28	119	120	20004
29	120	111	10016
30	121	1111	10017
31	122	443	10018
32	123	232	10019
33	124	234	10020
34	125	355	10021
35	126	17	10022
100	112	3341	10012
101	113	1	10013
102	114	0	10014
103	115	995	10015

MAINTENANCERECORD1

MAINTENANCEPERFORMED	NUMBEROFWORKERS
Oil Change	2
Brake Inspection	3
Structural Check	4
Electrical Test	2
Safety Audit	5

MAINTENANCERECORD2

RIDENAME	RECORDID	MAINTENANCEPERFORMED	MAINTENANCEDATE
RollerCoaster	1	Oil Change	2025-01-10 00:00:00
FerrisWheel	1	Brake Inspection	2025-02-20 00:00:00
HauntedHouse	1	Structural Check	2025-03-05 00:00:00
LogFlume	1	Electrical Test	2025-04-11 00:00:00
BumperCars	1	Safety Audit	2025-05-22 00:00:00

OFFERSROOMTYPE

HOTELNAME	ROOMNAME	QUANTITY
OceanView	Single	20
MountainInn	Double	15
CityCenter	Suite	5
LakesideLodge	Cabin	8
HarborHotel	Penthouse	2

RATEMODIFIER

RATECODE	MODIFIER
0	1
1	0.9
2	1.1
3	0.8
4	1.2

RIDE

RIDENAME	CAPACITY	INSTALLDATE
RollerCoaster	40	2015-06-01 00:00:00
FerrisWheel	30	2010-04-12 00:00:00
HauntedHouse	20	2018-10-10 00:00:00
LogFlume	25	2012-07-21 00:00:00
BumperCars	15	2011-09-14 00:00:00

ROOMTYPE1

CATEGORY	BASERATE
Economy	100
Standard	150
Luxury	300
Premium	500
Rustic	120

ROOMTYPE2

ROOMNAME	MAXGUESTS	CATEGORY
Single	1	Economy
Double	2	Standard
Suite	4	Luxury
Penthouse	6	Premium
Cabin	3	Rustic

SEASONPASS1

SEASONPASSLEVEL	SEASONSTART	SEASONEND
Gold	2025-01-01 00:00:00	2025-12-31 00:00:00
Silver	2025-04-01 00:00:00	2025-03-31 00:00:00
Bronze	2025-06-01 00:00:00	2025-05-31 00:00:00
Platinum	2025-02-01 00:00:00	2025-01-31 00:00:00
Member	2025-03-01 00:00:00	2025-02-28 00:00:00

SEASONPASS2

UUID	SEASONPASSLEVEL	SEASONSTART	SEASONEND	LOYALTYID
9001	Gold	2025-01-01 00:00:00	2025-12-31 00:00:00	101
9002	Silver	2025-04-01 00:00:00	2025-03-31 00:00:00	102
9003	Bronze	2025-06-01 00:00:00	2025-05-31 00:00:00	103
9004	Platinum	2025-02-01 00:00:00	2025-01-31 00:00:00	104
9005	Member	2025-03-01 00:00:00	2025-02-28 00:00:00	105

SEASONPASSSPECIAL

UUID	RATECODE
9001	1
9002	2
9003	3
9004	4
9005	0

TICKET1

VALIDFROM	VALIDHOURS	VALIDUNTIL
2025-07-01 00:00:00	8	2025-07-31 00:00:00
2025-07-10 00:00:00	8	2025-07-31 00:00:00
2025-07-11 00:00:00	8	2025-07-31 00:00:00
2025-07-12 00:00:00	8	2025-07-31 00:00:00
2025-08-01 00:00:00	10	2025-08-31 00:00:00
2025-09-01 00:00:00	6	2025-09-30 00:00:00
2025-09-14 00:00:00	8	2025-09-30 00:00:00
2025-09-14 00:00:00	9	2025-09-30 00:00:00
2025-10-01 00:00:00	12	2025-10-31 00:00:00
2025-11-01 00:00:00	4	2025-11-30 00:00:00

TICKET2

TICKETID	BOOKINGNUMBER	VALIDFROM	REMAININGUSES	VALIDHOURS
3333	1002	2025-08-01 00:00:00	2	10
3334	1002	2025-08-01 00:00:00	1	10
5001	1001	2025-07-10 00:00:00	5	8
5002	1002	2025-08-01 00:00:00	3	10
5003	1003	2025-09-01 00:00:00	2	6
5004	1004	2025-10-01 00:00:00	4	12
5005	1005	2025-11-01 00:00:00	1	4
5112	1111	2025-07-10 00:00:00	5	8
5113	1112	2025-07-10 00:00:00	5	8
5114	1113	2025-07-11 00:00:00	5	8
5115	1114	2025-07-12 00:00:00	5	8
5116	1115	2025-07-12 00:00:00	5	8
6666	2223	2025-09-14 00:00:00	6	8
6667	2224	2025-09-14 00:00:00	7	8
6668	2225	2025-09-14 00:00:00	2	8
6669	2226	2025-09-14 00:00:00	4	9
6670	2227	2025-09-14 00:00:00	4	8
8886	2323	2025-09-14 00:00:00	1	8
8887	2424	2025-09-14 00:00:00	4	8
8888	2525	2025-09-14 00:00:00	3	8
8889	2626	2025-09-14 00:00:00	7	8
8890	2727	2025-09-14 00:00:00	4	8
63203	7433	2025-09-01 00:00:00	2	6
63204	7434	2025-09-01 00:00:00	4	6
63205	7435	2025-09-01 00:00:00	1	6
63206	7436	2025-09-01 00:00:00	2	6
63207	7437	2025-09-01 00:00:00	4	6
63208	7438	2025-09-01 00:00:00	1	6